Networks Neural Physics-Informed

Problem Instability Stream Two Solving For Framework A

Shetaie Hosein

۱۴۰۴ Aban

تست فونت فارسی

- این یک اسلاید آزمایشی برای بررسی نمایش درست متن فارسی است
- اعداد فارسی: ۰۱۲۳۴۵۶۷۸۹
- ترکیب فارسی و لاتین: Vazirmatn و xepersian

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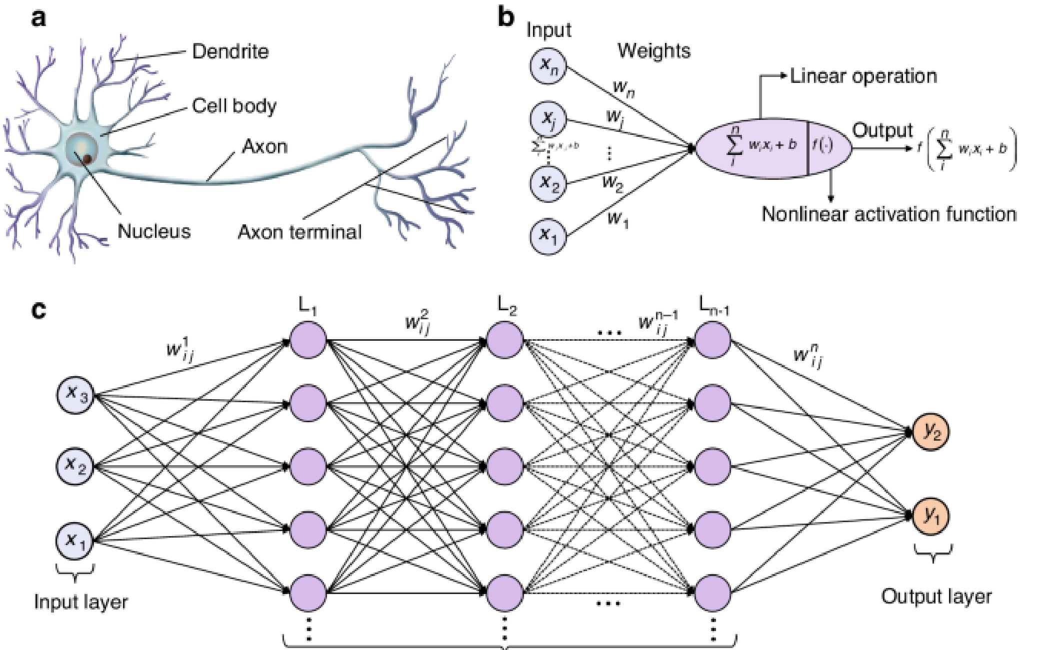
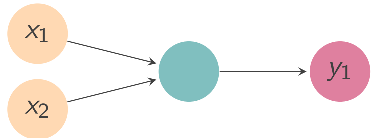

a
Dendrite
Cell body
Axon
Nucleus
Axon terminal
b
Input
Weights
Linear operation
Output
Nonlinear activation function
c
Input layer
Output layer

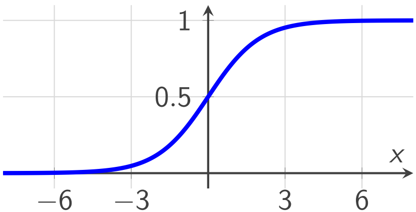
logic AND for Networks

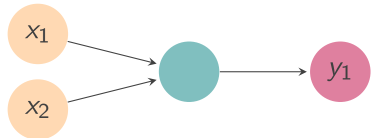


$$y_1 = f(w_1x_1 + w_2x_2 + b)$$

$y_1 = x_1 \wedge x_2$	x_2	x_1
۰	۰	۰
۰	۱	۰
۰	۰	۱
۱	۱	۱

$$f(x) = \frac{1}{1 + e^{-x}}$$





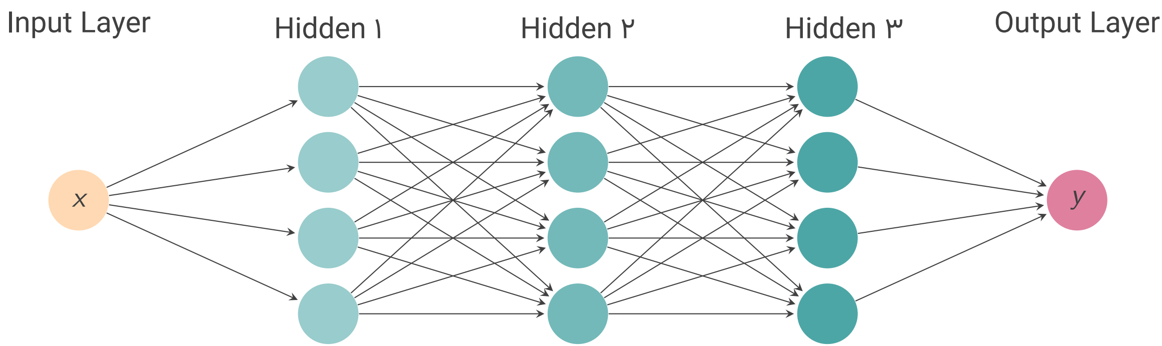
$$y_1 = f(w_1 x_1 + w_2 x_2 + b)$$

$y_1 = x_1 \wedge x_2$	x_2	x_1
۰	۰	۰
۰	۱	۰
۰	۰	۱
۱	۱	۱

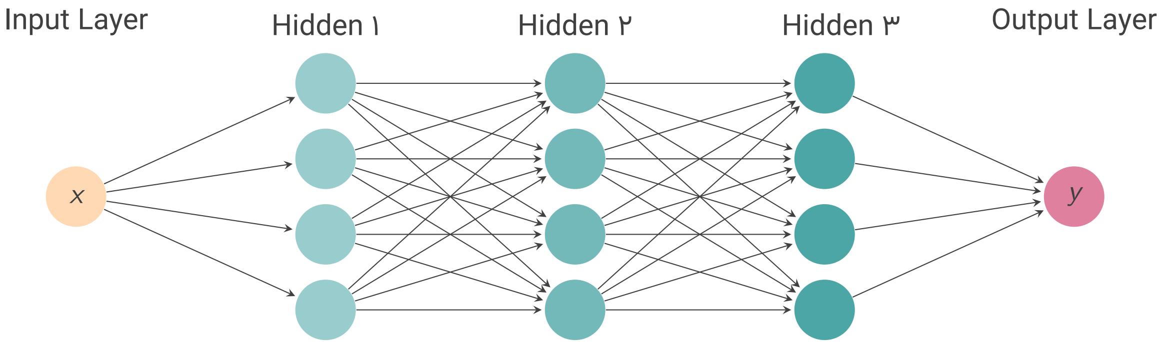
$$y_1 = f(5.5x_1 + 5.5x_2 - 8)$$

Output	f(sum)	Sum Weighted	x_2	x_1
۰	۰۰۰۳.۰	۰.۸-	۰	۰
۰	۰۷۵.۰	۵.۲-	۱	۰
۰	۰۷۵.۰	۵.۲-	۰	۱
۱	۹۵۲.۰	۰.۳	۱	۱

جدول: operation. AND for output network Neural

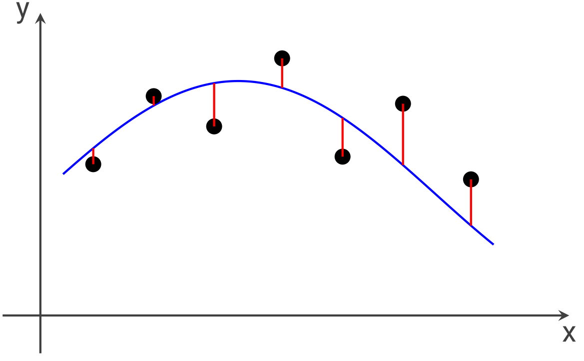


Input Layer
Hidden ۱
Hidden ۲
Hidden ۳
Output Layer
x
y

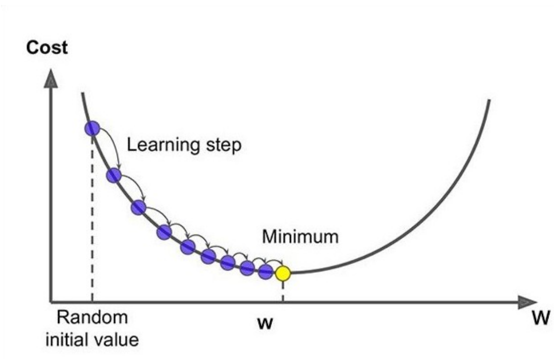


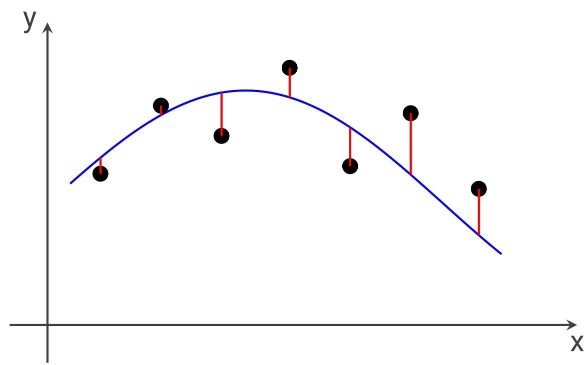
$$\text{Loss} = \frac{1}{N}\sum_{i}^{N}\left(y_{\text{nn}}(x_i) - y_{\text{true}}(x_i)\right)^2$$

$$\text{Cost} = \frac{1}{N}\sum_{i}^{N}\left(y_{\text{nn}}(x_i) - y_{\text{true}}(x_i)\right)^2$$



$$\text{Cost} = \frac{1}{N}\sum_{i}^{N}\left(y_{\text{nn}}(x_i) - y_{\text{true}}(x_i)\right)^2$$





Contents of Table

Learning Physics-Driven to Data-Driven From

Models Data-Driven Purely of Limitations The

- datasets high-quality large، Require
- noisy or scarce are data when poorly Perform
- momentum) energy، mass، of conservation (e.g.، laws physical violate May

Learning Physics-Guided for Motivation

- known are laws but limited are data problems، scientific In
- learning guide to constraints) (PDEs، knowledge domain Use
- function loss the into directly physics Integrate
- alone data of instead physics + data both from Learn

ELSEVIER

Journal of Computational Physics

Volume 378, 1 February 2019, Pages 686-707

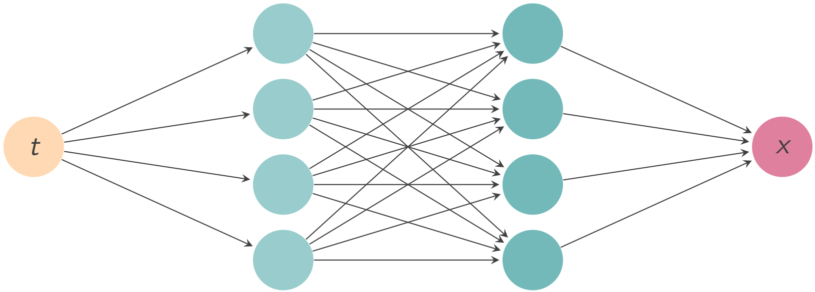
Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations

M. Raissi [a], P. Perdikaris [b], G.E. Karniadakis [a]

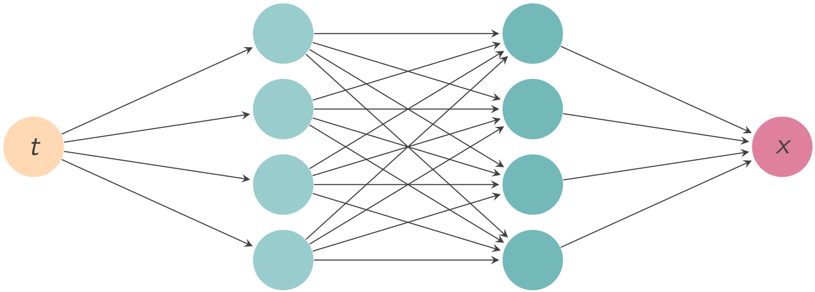
https://doi.org/10.1016/j.jcp.2018.10.045 ↗

Highlights

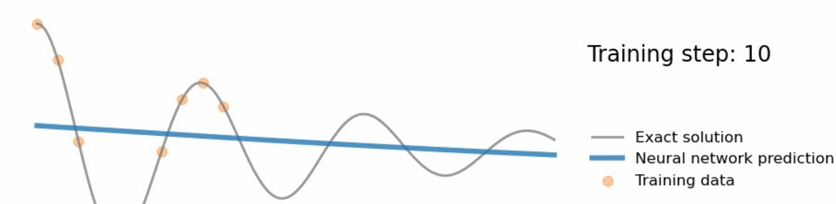
- We put forth a deep learning framework that enables the synergistic combination of mathematical models and data.
- We introduce an effective mechanism for regularizing the training



$$\text{Loss} = \frac{1}{N}\sum_{i}^{N}\left(x_{\text{nn}}(t_i) - x_{\text{true}}(t_i)\right)^2$$

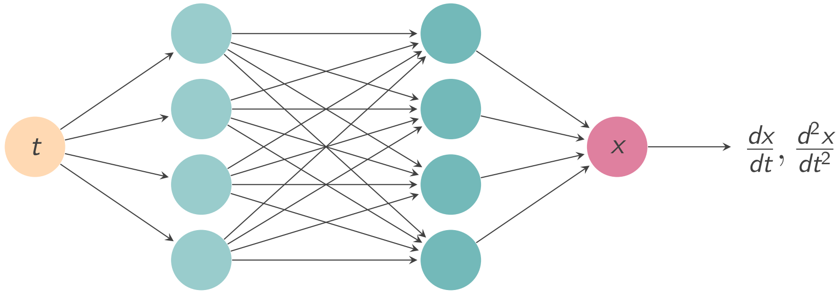


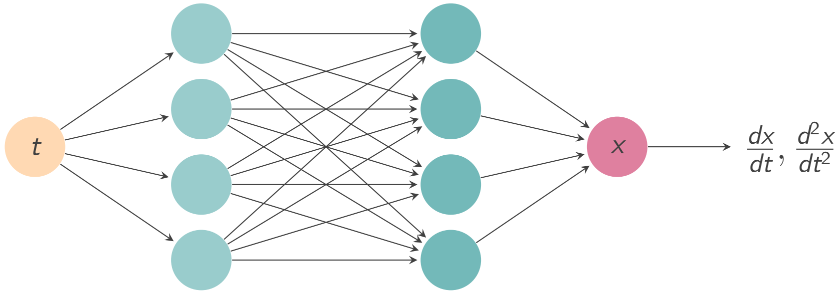
$$\text{Loss} = \frac{1}{N}\sum_{i}^{N}\left(x_{\text{nn}}(t_i) - x_{\text{true}}(t_i)\right)^2$$



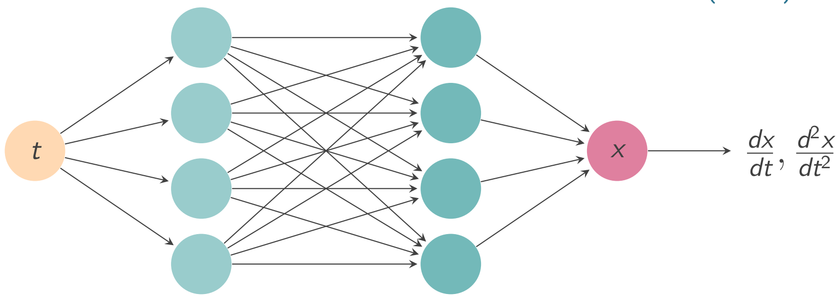
Oscillator

$$m\frac{d^2x}{dt^2}+\mu\frac{dx}{dt}+kx=0$$

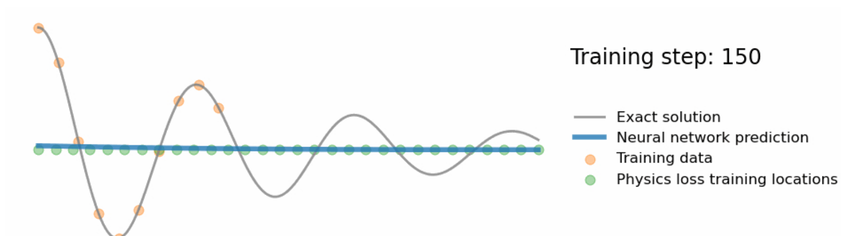

t
x
$\frac{dx}{dt}, \frac{d^2x}{dt^2}$



$$\text{Loss} = \frac{1}{N}\sum_{i}^{N}\left(x_{\text{nn}}(t_i) - x_{\text{true}}(t_i)\right)^2 + \frac{1}{M}\sum_{j}^{M}\left(\left[m\frac{d^2}{dt^2} + \mu\frac{d}{dt} + k\right]x_{\text{nn}}(t_j)\right)^2$$



$$\text{Loss} = \frac{1}{N}\sum_{i}^{N}\left(x_{\text{nn}}(t_i) - x_{\text{true}}(t_i)\right)^2 + \frac{1}{M}\sum_{j}^{M}\left(\left[m\frac{d^2}{dt^2} + \mu\frac{d}{dt} + k\right]x_{\text{nn}}(t_j)\right)^2$$



problem Forward

the and conditions boundary or and initial the only knowing by equation differential a Solving

problem Forward

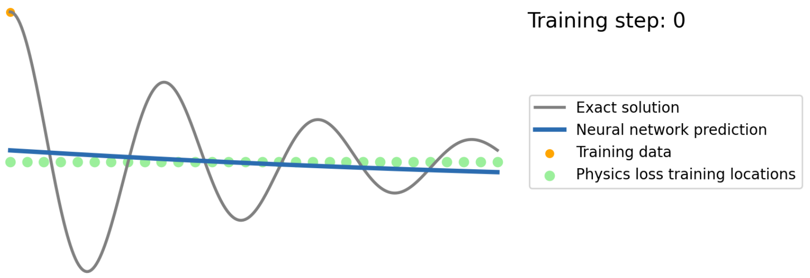
the and conditions boundary or and initial the only knowing by equation differential a Solving data. indirect or partial from them infer to is goal

$$\text{Loss} = (x_{\text{nn}}(t=0)-1)^2 + \lambda_1\left(\frac{dx_{\text{nn}}}{dt}(t=0)\right)^2 + \frac{\lambda_2}{N}\sum_j^N\left(\left[m\frac{d^2}{dt^2}+\mu\frac{d}{dt}+k\right]x_{\text{nn}}(t_j)\right)^2$$

problem Forward

the and conditions boundary or and initial the only knowing by equation differential a Solving data. indirect or partial from them infer to is goal

$$\text{Loss} = (x_{\text{nn}}(t=0)-1)^2 + \lambda_1\left(\frac{dx_{\text{nn}}}{dt}(t=0)\right)^2 + \frac{\lambda_2}{N}\sum_j^N\left(\left[m\frac{d^2}{dt^2}+\mu\frac{d}{dt}+k\right]x_{\text{nn}}(t_j)\right)^2$$



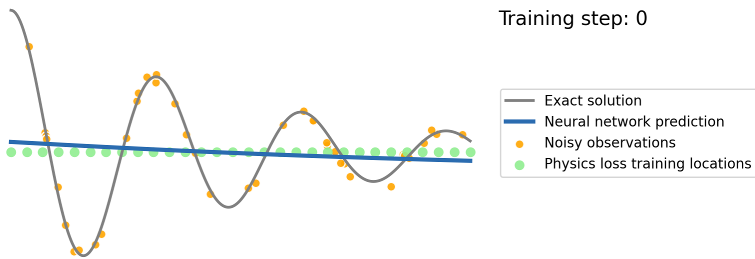
the and unknown are it of parts some but known، is form equation the problem، inverse an In the and unknown are it of parts some but known، is form equation the problem، inverse an In data. indirect or partial from them infer to is goal

$$\text{Loss} = \frac{\lambda}{N}\sum_{i}^{N}\left(x_{\text{nn}}(t_i) - x_{\text{true}}(t_i)\right)^2 + \frac{1}{M}\sum_{j}^{M}\left(\left[m\frac{d^2}{dt^2} + \mu\frac{d}{dt} + k\right]x_{\text{nn}}(t_j)\right)^2$$

problem Inverse

the and unknown are it of parts some but known، is form equation the problem، inverse an In
data. indirect or partial from them infer to is goal

$$\text{Loss} = \frac{\lambda}{N}\sum_{i}^{N}\left(x_{\text{nn}}(t_i) - x_{\text{true}}(t_i)\right)^2 + \frac{1}{M}\sum_{j}^{M}\left(\left[m\frac{d^2}{dt^2} + \mu\frac{d}{dt} + k\right]x_{\text{nn}}(t_j)\right)^2$$



PINN for solving Vlasov–Poisson equation

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Physics-informed neural networks for solving forward and inverse Vlasov–Poisson equation via fully kinetic simulation

Baiyi Zhang, Guobiao Cai, Huiyan Weng, Weizong Wang, Lihui Liu and Bijiao He

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DOI 10.1088/2632-2153/ad03d5

Article PDF

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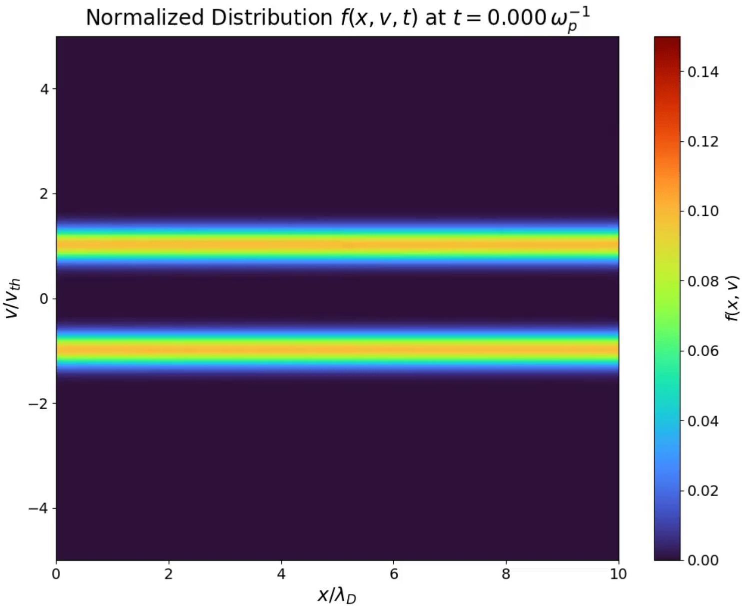


Normalized Distribution $f(x, v, t)$ at $t = 0.000\,\omega_p^{-1}$
v/v_{th}
x/λ_D
$f(x, v)$
4
2
0
−2
−4
0
2
4
6
8
10
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00

equation: Vlasov-Poisson

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f + \frac{q}{m}(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \frac{\partial f}{\partial \mathbf{v}} = 0$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon}$$

Density:

equation: Vlasov-Poisson

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f + \frac{q}{m}(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \frac{\partial f}{\partial \mathbf{v}} = 0$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon}$$

Density:

$$\rho = e(n_e - Z_i n_i)$$

$$n_{\mathrm{e}} = \int_{-\infty}^{+\infty} f(t, x, v)\, \mathrm{d}v$$

equation: Vlasov-Poisson ۱D

equation: Vlasov-Poisson

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f + \frac{q}{m} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \frac{\partial f}{\partial \mathbf{v}} = 0$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon}$$

Density:

$$\rho = e(n_e - Z_i n_i)$$

$$n_{\mathrm{e}} = \int_{-\infty}^{+\infty} f(t, x, v) \, \mathrm{d}v$$

equation: Vlasov-Poisson 1D

$$\frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} + \frac{eE}{m_{\mathrm{e}}} \frac{\partial f}{\partial v} = 0$$

$$\frac{\partial E}{\partial x} = \frac{e(n_{\mathrm{e}} - n_i)}{\varepsilon}$$

Normalization:

$$x = \frac{x^*}{\lambda_\mathrm{D}},\ v = \frac{v^*}{v_\mathrm{th}},\ t = \frac{t^*}{\omega_\mathrm{p}^{-1}},\ m = \frac{m^*}{m_\mathrm{e}},\ q = \frac{q^*}{e},\ E = E^* \frac{e\lambda_\mathrm{D}}{kT_\mathrm{e}}$$

Normalization:

$$x = \frac{x^*}{\lambda_\mathrm{D}},\; v = \frac{v^*}{v_\mathrm{th}},\; t = \frac{t^*}{\omega_\mathrm{p}^{-1}},\; m = \frac{m^*}{m_\mathrm{e}},\; q = \frac{q^*}{e},\; E = E^* \frac{e\lambda_\mathrm{D}}{kT_\mathrm{e}}$$

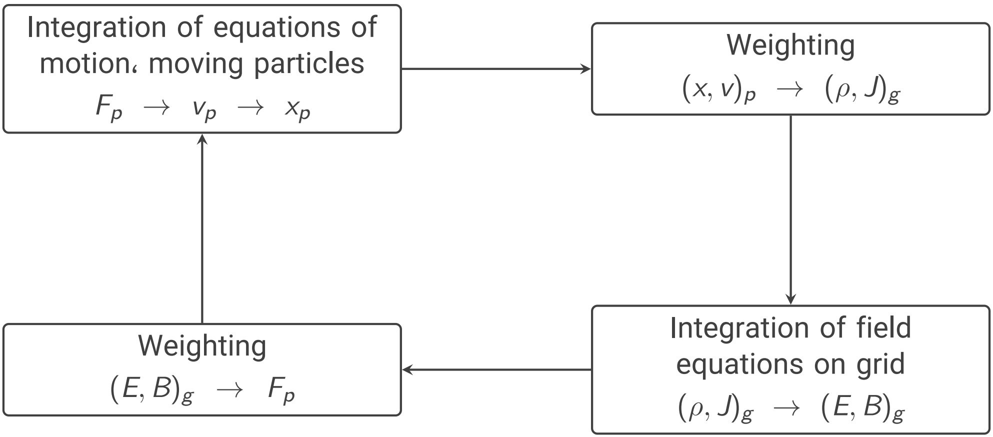
equation: Vlasov-Poisson ۱D Normalized

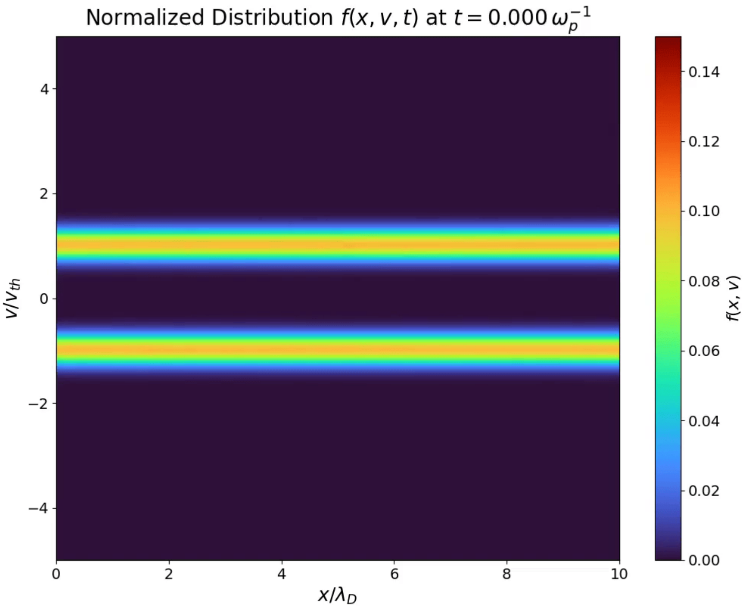
$$\frac{\partial f}{\partial t} + v\frac{\partial f}{\partial x} - \mathrm{E}\frac{\partial f}{\partial v} = 0$$

$$\frac{\partial \mathrm{E}}{\partial x} = \int_{-\infty}^{+\infty} f(t, x, v)\,\mathrm{d}v - 1$$

simulation PIC

Instability Stream Two ۳


Integration of equations of motion، moving particles
$F_p \rightarrow v_p \rightarrow x_p$
Weighting
$(x, v)_p \rightarrow (\rho, J)_g$
Integration of field equations on grid
$(\rho, J)_g \rightarrow (E, B)_g$
Weighting
$(E, B)_g \rightarrow F_p$



Normalized Distribution $f(x, v, t)$ at $t = 0.000\,\omega_p^{-1}$
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4
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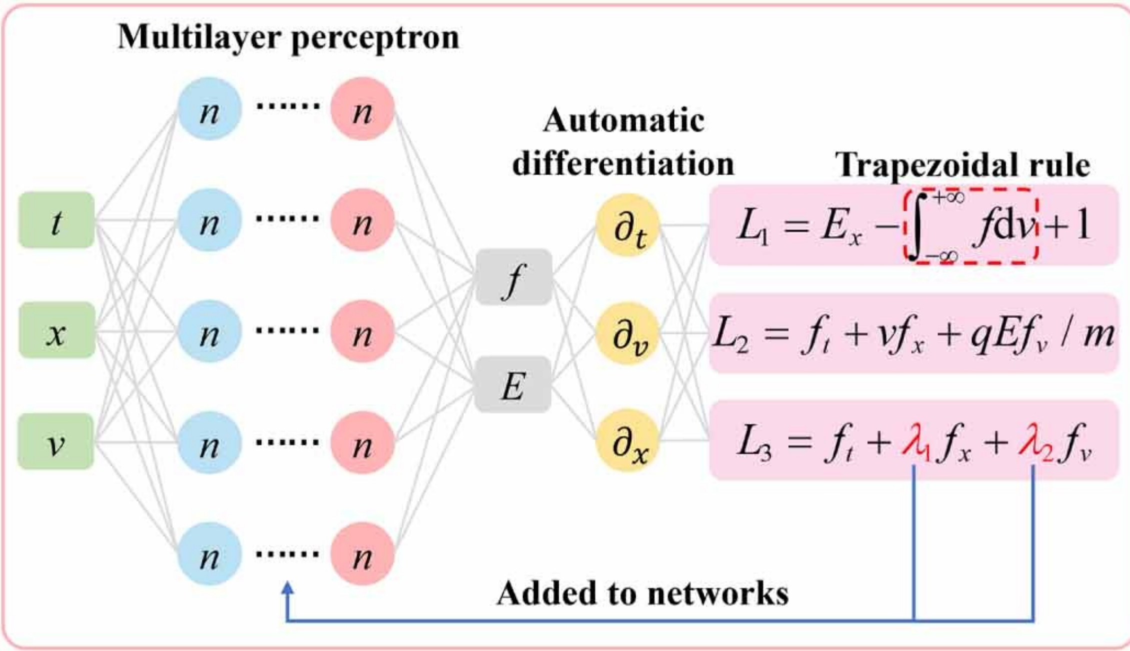
0.04
0.02
0.00

Contents of Table

equation Vlasov–Poisson the solving for PINNs



PINN-Vlasov Forward

$$L = L_{\text{eq}} + L_{\text{IC}} + L_{\text{BC}}$$

$$L = L_{\mathrm{eq}} + L_{\mathrm{IC}} + L_{\mathrm{BC}}$$

$$L_{\mathrm{eq}} = \frac{1}{N_{\mathrm{eq}}} \sum_{n=1}^{N_{\mathrm{eq}}} \left[\left(\frac{\partial f}{\partial t} + v\frac{\partial f}{\partial x} + \frac{qE}{m}\frac{\partial f}{\partial v} \right)^2 + \left(\frac{\partial E}{\partial x} - \int_{-\infty}^{+\infty} f \mathrm{d}v + 1 \right)^2 \right]_n$$

$$L_{\mathrm{IC}} = \frac{1}{N_{\mathrm{IC}}} \sum_{n=1}^{N_{\mathrm{IC}}} \left[\left(f_{\mathrm{IC}}^{\mathrm{r}} - f_{\mathrm{IC}}^{\mathrm{t}} \right)^2 \right]_n$$

$$L_{\mathrm{BC}} = \frac{1}{N_{\mathrm{BC}}} \sum_{n=1}^{N_{\mathrm{BC}}} \left[\left(f_{\mathrm{BC}}^{\mathrm{r}} - f_{\mathrm{BC}}^{\mathrm{t}} \right)^2 \right]_n$$

$$L = L^{'}_{\mathrm{eq}} + L_{\mathrm{f}}$$

$$L = L'_{\mathrm{eq}} + L_{\mathrm{f}}$$

$$L'_{\mathrm{eq}} = \frac{1}{N'_{\mathrm{eq}}} \sum_{n=1}^{N'_{\mathrm{eq}}} \left[\left(\frac{\partial f}{\partial t} + \lambda_1 v \frac{\partial f}{\partial x} + \lambda_2 E \frac{\partial f}{\partial v} \right)^2 + \left(\frac{\partial E}{\partial x} - \int_{-\infty}^{+\infty} f \mathrm{d}v + 1 \right)^2 \right]_n$$

$$L_{\mathrm{f}} = \frac{1}{N_{\mathrm{f}}} \sum_{n=1}^{N_{\mathrm{f}}} \left[\left(f^{\mathrm{r}} - f^{\mathrm{t}} \right)^2 \right]_n$$

$$L = L'_{\text{eq}} + L_{\text{f}}$$

$$L'_{\text{eq}} = \frac{1}{N'_{\text{eq}}} \sum_{n=1}^{N'_{\text{eq}}} \left[\left(\frac{\partial f}{\partial t} + \lambda_1 v \frac{\partial f}{\partial x} + \lambda_2 E \frac{\partial f}{\partial v} \right)^2 + \left(\frac{\partial E}{\partial x} - \int_{-\infty}^{+\infty} f \mathrm{d}v + 1 \right)^2 \right]_n$$

$$L_{\text{f}} = \frac{1}{N_{\text{f}}} \sum_{n=1}^{N_{\text{f}}} \left[\left(f^{\text{r}} - f^{\text{t}} \right)^2 \right]_n$$

$$\lambda_1 = 1.0$$

$$\lambda_2 = \frac{q}{m} = \frac{-1.0}{1.0} = -1.0$$

Challenges

- field electric the paper، the In **problem forward in model the to provided field Electric**
the limit may this training. forward during PINN the to supplied directly was $E(x, t)$
physics. the from self-consistently E infer to ability model's

- requires $\frac{\partial E}{\partial x} = \int f dv - 1$ term Poisson The **equation Poisson the in Integration**
affects rule) trapezoidal (e.g.، method integration of choice The integration. numerical
accuracy. and stability

- term loss each of importance relative The **components loss of weighting Unknown**
bias or convergence hinder can weighting defined-improper clearly not is $(L_{\text{eq}}, L_{\text{IC}}, L_{\text{BC}})$
network. the

- training how on depends strongly performance The **strategy resampling and Sampling**
intervals resampling and frequency sampling Optimal .(t, x, v) in chosen are points
questions. research open remain

- **Electric field provided to the model in forward problem** In the paper، the electric field $E(x, t)$ was directly supplied to the PINN during forward training. This may limit the model's ability to infer E self-consistently from the physics.

- **Integration in the Poisson equation** The Poisson term $\frac{\partial E}{\partial x} = \int f\,dv - 1$ requires numerical integration. The choice of integration method (e.g.، trapezoidal rule) affects stability and accuracy.

- **Unknown weighting of loss components** The relative importance of each loss term $(L_{\text{eq}}, L_{\text{IC}}, L_{\text{BC}})$ is not clearly defined–improper weighting can hinder convergence or bias the network.

- **Sampling and resampling strategy** The performance strongly depends on how training points are chosen in (t, x, v). Optimal sampling frequency and resampling intervals remain open research questions.

Observation

Balancing physical accuracy، numerical stability، and training efficiency is still the main challenge for Vlasov-PINN frameworks.

- and learning deep between gap the bridge **(PINNs) Networks Neural Physics-Informed** modeling. physical
- consistency. physical ensure they function، loss the into **PDEs governing** embedding By
- high with **problems inverse and forward** both solving for capability Demonstrated interpretability.

Work Future

Conclusion ۵

- PDEs. complex for training faster — **efficiency computational Improving**
- points. collocation of selection dynamic — **sampling Adaptive**
- dimensions higher collisions، fields، magnetic include — **extension Multi-physics** (۲D/۳D).

Work Future

Conclusion ۵

- **efficiency computational Improving** — faster training for complex PDEs.
- **sampling Adaptive** — dynamic selection of collocation points.
- **Multi-physics extension** — include magnetic fields، collisions، higher dimensions (۲D/۳D).

Future PINNs = Physics + Data + Efficiency Scientific AI.

Q&A

listening! for you Thank